Q1.

MyTest.java | Q1Employee.java | Q21Uni.java | newTest2.java

```java
package assignment;

public class Q1Employee {

    private int empId;
    private String empName;
    private double salary;

    public int getEmpId() {
        return empId;
    }

    public void setEmpId(int empId) {
        this.empId = empId;
    }

    public String getEmpName() {
        return empName;
    }

    public void setEmpName(String empName) {
        this.empName = empName;
    }

    public double getSalary() {
        return salary;
    }

    public void setSalary(double salary) {
        this.salary = salary;
    }

    void displayDetails() {
        System.out.println("Employee id is: "+ empId);
        System.out.println("Employee name is: "+ empName);
```

<terminated> Q1Employee [Java Applicat

```
Employee id is: 101
Employee name is: Rahul
Employee salary is: 10000.0
```

MyTest.java | Q1Employee.java | Q21Uni.java | newTest2.java

```java
        return empName;
    }

    public void setEmpName(String empName) {
        this.empName = empName;
    }

    public double getSalary() {
        return salary;
    }

    public void setSalary(double salary) {
        this.salary = salary;
    }

    void displayDetails() {
        System.out.println("Employee id is: "+ empId);
        System.out.println("Employee name is: "+ empName);
        System.out.println("Employee salary is: "+ salary);

    }

    public static void main(String[] args) {

        Q1Employee obj = new Q1Employee();
        obj.setEmpId(101);
        obj.setEmpName("Rahul");
        obj.setSalary(10000.00);

        obj.displayDetails();

    }

}
```

<terminated> Q1Employee [Java Application

```
Employee id is: 101
Employee name is: Rahul
Employee salary is: 10000.0
```

Q2.

MyTest.java Q1Employee.java Q2Vehicle.java Q21Uni.java newTest2.java

```java
package assignment;

class Vehicle
{
    void fuelType()
    {
        System.out.println("Runs on fuel");
    }
}

class ElectricCar extends Vehicle
{
    void fuelType()
    {
        System.out.println("Runs on electricity");
    }
}

public class Q2Vehicle {

    public static void main(String[] args) {

        Vehicle obj = new Vehicle();
        obj.fuelType();

        ElectricCar obj1 = new ElectricCar();
        obj1.fuelType();

    }

}
```

<terminated> Q2Vehicle [Java Application] C

```
Runs on fuel
Runs on electricity
```

Q3.

Q1Employee.java Q2Vehicle.java Q3Product.java Q21Uni.java

```java
package assignment;

public class Q3Product {

    int productId;
    String productName;
    float price;

    Q3Product()
    {
      System.out.println("Product Created");
    }

    Q3Product(int pId, String pName, float pPrice)
    {
      this.productId=pId;
      this.productName=pName;
      this.price=pPrice;
    }

    void displayProduct()
    {
        System.out.println("Product id is: "+productId);
        System.out.println("Product name is: "+productName);
        System.out.println("Product price is: "+price);
    }

    public static void main(String[] args) {

        Q3Product obj = new Q3Product();
        Q3Product obj1 = new Q3Product(23211, "TV", 23999);
        obj1.displayProduct();

    }
}
```

<terminated> Q3Product [Java Applicatior

```
Product Created
Product id is: 23211
Product name is: TV
Product price is: 23999.0
```

Q4.

```java
package assignment;

public class Q4Calculator {

    int add(int a, int b)
    {
        return a+b;
    }

    double add(double a, double b)
    {
        return a+b;
    }

    public static void main(String[] args) {

      Q4Calculator obj = new Q4Calculator();
      System.out.println(obj.add(10, 7));
      System.out.println(obj.add(12.7, 22));

    }

}
```

<terminated> Q4Calculator [Java Applicatio

```
17
34.7
```

Q5.

```java
package assignment;

abstract class Animal{

    abstract void sound();
}

class Dog extends Animal
{
    void sound()
    {
        System.out.println("Bow bow");
    }
}

class Cat extends Animal
{
    void sound()
    {
        System.out.println("Mewo mewo");
    }
}

public class Q5Animal {

    public static void main(String[] args) {

        Dog d = new Dog();
        d.sound();
        Cat c = new Cat();
        c.sound();

    }
}
```

<terminated> Q5Animal [Java Application]

```
Bow bow
Mewo mewo
```

Q6.

```java
package assignment;

interface Payment
{
    void makePayment();
}

class UPI implements Payment
{
    public void makePayment()
    {
        System.out.println("This is UPI payment");
    }
}

class CreditCard implements Payment
{
    public void makePayment()
    {
        System.out.println("This is CreditCard payment");
    }
}

public class Q6Payment {

    public static void main(String[] args) {

        Payment pay = new UPI();
        pay.makePayment();

        Payment pay1 = new CreditCard();
        pay1.makePayment();

    }
```

<terminated> Q6Payment [Java Application]

```
This is UPI payment
This is CreditCard payment
```

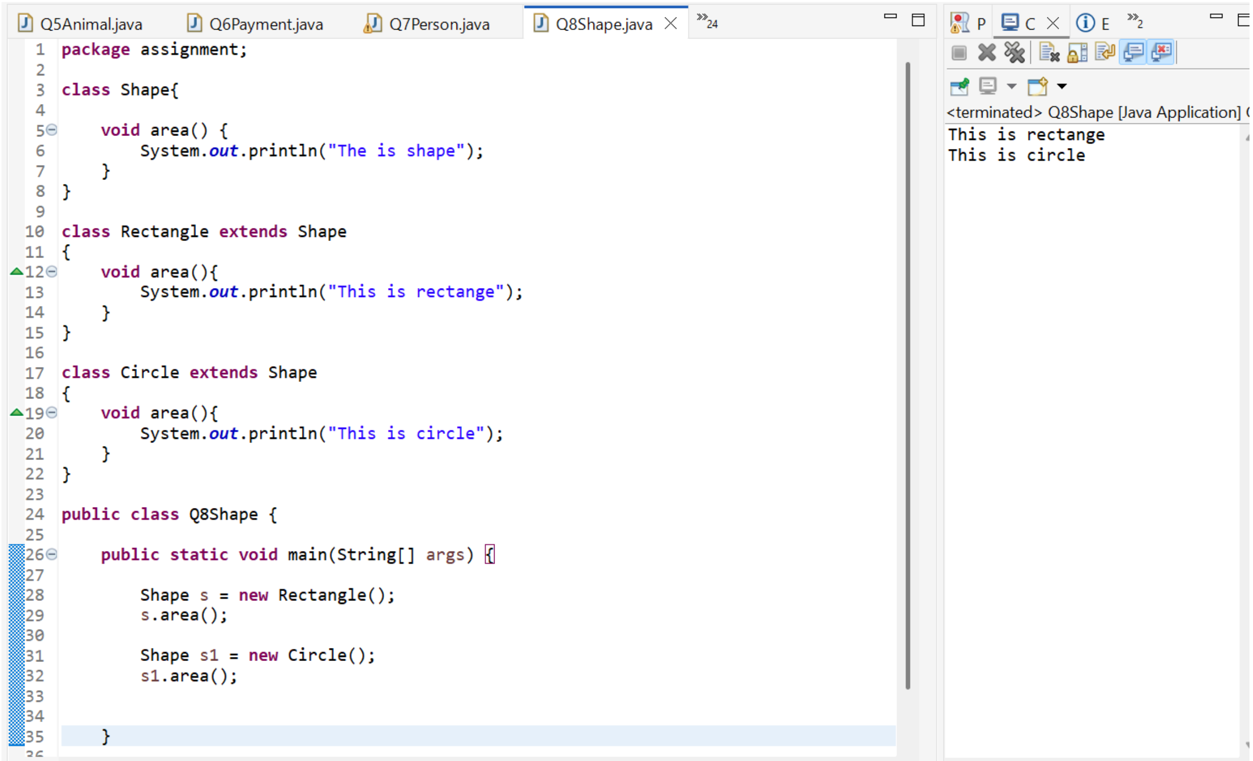
Q7.

```java
package assignment;

class Person{

    Person()
    {
        System.out.println("Person Created");
    }
}

class Student extends Person
{
    Student()
    {
        super();
        System.out.println("Student Created");
    }
}

public class Q7Person {

    public static void main(String[] args) {

        Student st = new Student();

    }
}
```

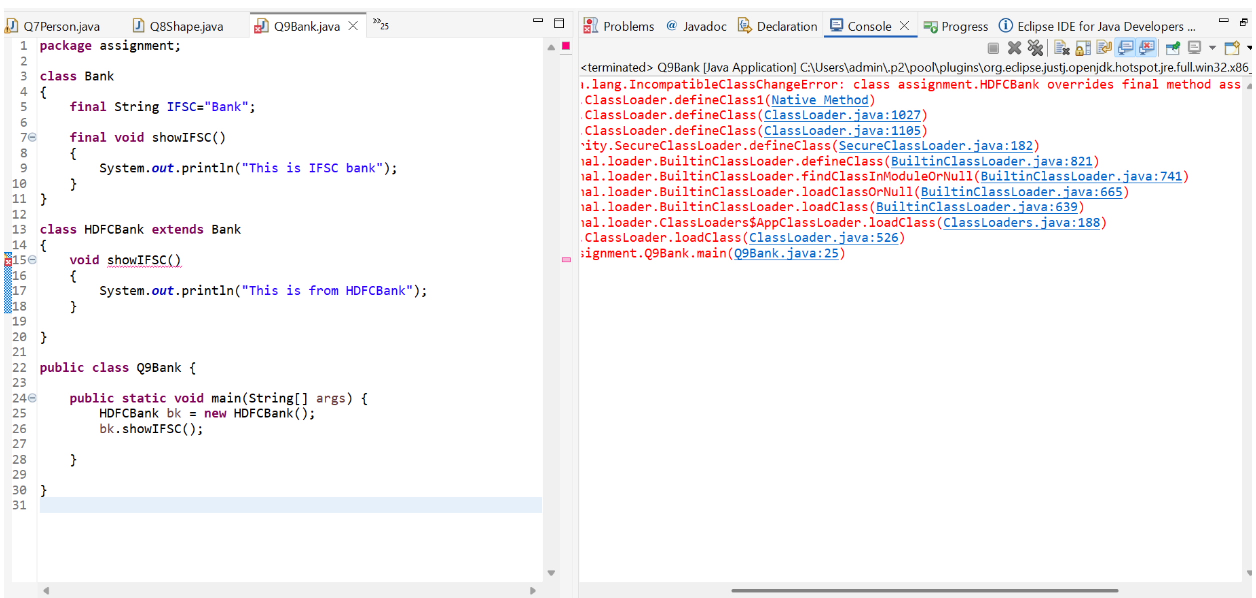
<terminated> Q7Person [Java Application

```
Person Created
Student Created
```

Q8.


Q5Animal.java
Q6Payment.java
Q7Person.java
Q8Shape.java
package assignment;
class Shape{
void area() {
System.out.println("The is shape");
}
}
class Rectangle extends Shape
{
void area(){
System.out.println("This is rectange");
}
}
class Circle extends Shape
{
void area(){
System.out.println("This is circle");
}
}
public class Q8Shape {
public static void main(String[] args) {
Shape s = new Rectangle();
s.area();
Shape s1 = new Circle();
s1.area();
}
<terminated> Q8Shape [Java Application]
This is rectange
This is circle

Q9.


Q7Person.java
Q8Shape.java
Q9Bank.java
package assignment;
class Bank
{
final String IFSC="Bank";
final void showIFSC()
{
System.out.println("This is IFSC bank");
}
}
class HDFCBank extends Bank
{
void showIFSC()
{
System.out.println("This is from HDFCBank");
}
}
public class Q9Bank {
public static void main(String[] args) {
HDFCBank bk = new HDFCBank();
bk.showIFSC();
}
}
Problems
Javadoc
Declaration
Console
Progress
Eclipse IDE for Java Developers ...
<terminated> Q9Bank [Java Application] C:\Users\admin\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_
.lang.IncompatibleClassChangeError: class assignment.HDFCBank overrides final method ass
.ClassLoader.defineClass1(Native Method)
.ClassLoader.defineClass(ClassLoader.java:1027)
.ClassLoader.defineClass(ClassLoader.java:1105)
ity.SecureClassLoader.defineClass(SecureClassLoader.java:182)
al.loader.BuiltinClassLoader.defineClass(BuiltinClassLoader.java:821)
al.loader.BuiltinClassLoader.findClassInModuleOrNull(BuiltinClassLoader.java:741)
al.loader.BuiltinClassLoader.loadClassOrNull(BuiltinClassLoader.java:665)
al.loader.BuiltinClassLoader.loadClass(BuiltinClassLoader.java:639)
al.loader.ClassLoaders$AppClassLoader.loadClass(ClassLoaders.java:188)
.ClassLoader.loadClass(ClassLoader.java:526)
ignment.Q9Bank.main(Q9Bank.java:25)

Q10.

```java
package assignment;

class CStudent
{
    static String collegeName="Dayananda sagar";
    String name;
    int rollNo;

    CStudent(String name, int rollNo) {
        this.name = name;
        this.rollNo = rollNo;
    }

    void display()
    {
        System.out.println("College name is: "+collegeName);
        System.out.println("Student name is: "+name);
        System.out.println("Student roll number is: "+rollNo);
        System.out.println("----------------------");
    }

}

public class Q10Student {

    public static void main(String[] args) {

        CStudent s1 = new CStudent("Alice", 101);
        CStudent s2 = new CStudent("Bob", 102);
        CStudent s3 = new CStudent("Charlie", 103);

        s1.display();
        s2.display();
        s3.display();

        CStudent.collegeName = "XYZ College";
```

Writable | Smart Insert | 14 : 19 : 233

<terminated> Q10Student [Java Application] C:\Users\admin\.p2\pool\pl

```
College name is: Dayananda sagar
Student name is: Alice
Student roll number is: 101
----------------------
College name is: Dayananda sagar
Student name is: Bob
Student roll number is: 102
----------------------
College name is: Dayananda sagar
Student name is: Charlie
Student roll number is: 103
----------------------
After changing college name:
College name is: XYZ College
Student name is: Alice
Student roll number is: 101
----------------------
College name is: XYZ College
Student name is: Bob
Student roll number is: 102
----------------------
College name is: XYZ College
Student name is: Charlie
Student roll number is: 103
----------------------
```

Q11.

Q8Shape.java | Q9Bank.java | Q10Student.java | Q11Library.java | »24

```java
package assignment;

class Library
{
    String libraryName;

    Library()
    {
        System.out.println("Welcome to the Library!");
    }

    void showLocation()
    {
        System.out.println("This library is located in Mumbai");
    }
}

public class Q11Library {

    public static void main(String[] args) {

        Library l = new Library();
        l.showLocation();
    }

}
```

Pr... | Ja... | De... | Co... | Pr... | Ecl...

<terminated> Q11Library [Java Application] C:\Users\admin\.p2\pool\p

```
Welcome to the Library!
This library is located in Mumbai
```

Q12.

```java
package assignment;

class Account{
    private String accountHolderName;
    private float balance;

    public String getAccountHolderName() {
        return accountHolderName;
    }

    public void setAccountHolderName(String accountHolderName) {
        this.accountHolderName = accountHolderName;
    }

    public float getBalance() {
        return balance;
    }

    public void setBalance(float balance) {

        if (balance<0)
        {
        System.out.println("Please do not enter negative number");
        }
        else
        {
            this.balance = balance;
        }

    }

}

public class Q12Account {

    public static void main(String[] args) {
```

```
<terminated> Q12Account [Java Application] C:\Users\admin\.p2\pool\pl
Bharat BJ
9.0
```

Writable | Smart Insert | 42 : 45 : 816

Q9Bank.java | Q10Student.java | Q11Library.java | Q12Account.java | »24

Pr... | Ja... | De... | Co... | Pr... | Ecl...

```java
        this.accountHolderName = accountHolderName;
    }

    public float getBalance() {
        return balance;
    }

    public void setBalance(float balance) {

        if (balance<0)
        {
        System.out.println("Please do not enter negative number");
        }
        else
        {
            this.balance = balance;
        }

    }

}

public class Q12Account {

    public static void main(String[] args) {

        Account ac = new Account();
        ac.setAccountHolderName("Bharat BJ");
        ac.setBalance(9);
        System.out.println(ac.getAccountHolderName());
        System.out.println(ac.getBalance());

    }

}
```

```
<terminated> Q12Account [Java Application] C:\Users\admin\.p2\pool\pl
Bharat BJ
9.0
```

Writable | Smart Insert | 42 : 45 : 816

Q13.

Q10Student.java | Q11Library.java | Q12Account.java | Q13Inherit.java

```java
package assignment;

class Device{
    void start()
    {
        System.out.println("Device");
    }
}

class Mobile extends Device {
    void calling()
    {
        System.out.println("Mobile");
    }
}

class SmartPhone extends Mobile{
    void internet()
    {
        System.out.println("SmartPhone");
    }
  }

public class Q13Inherit {

    public static void main(String[] args) {

        SmartPhone sm = new SmartPhone();
        sm.internet();
        sm.calling();
        sm.start();
    }

}
```

<terminated> Q13Inherit [Java Application] C:\Users\admin\.p2\pool\plu

```
SmartPhone
Mobile
Device
```

Writable | Smart Insert | 23 : 1 : 295

Q14.

Q11Library.java | Q12Account.java | Q13Inherit.java | Q14HInherit....

```java
package assignment;

class Course{
    void courseInfo()
    {
        System.out.println("____Your course_____");
    }
}

class Science extends Course{
    void courseInfo()
    {
        System.out.println("Your course_____Science");
    }
}

class Commerce extends Course
{
    void courseInfo()
    {
        System.out.println("Your course_____Commerce");
    }
}
class Arts extends Course
{
    void courseInfo()
    {
        System.out.println("Your course_____Arts");
    }
}

public class Q14HInherit {

    public static void main(String[] args) {

        Course c = new Course();
```

<terminated> Q14HInherit [Java Application] C:\Users\admin\.p2\pool\pl

```
____Your course_____
Your course_____Science
Your course_____Commerce
Your course_____Arts
```

Writable | Smart Insert | 37 : 24 : 580

```java
        System.out.println("Your course_____Science");
    }
}

class Commerce extends Course
{
    void courseInfo()
    {
        System.out.println("Your course_____Commerce");
    }
}
class Arts extends Course
{
    void courseInfo()
    {
        System.out.println("Your course_____Arts");
    }
}

public class Q14HInherit {

    public static void main(String[] args) {

        Course c = new Course();
        c.courseInfo();
        Science s = new Science();
        s.courseInfo();
        Commerce com = new Commerce();
        com.courseInfo();
        Arts a = new Arts();
        a.courseInfo();

    }

}
```

```
<terminated> Q14HInherit [Java Application] C:\Users\admin\.p2\pool\pl
____Your course_____
Your course_____Science
Your course_____Commerce
Your course_____Arts
```

Q15.

```java
package assignment;

class LoanCalculator
{
    int amount;
    double interestRate;

    void calculateLoan(int amount)
    {
        this.amount = amount;
        System.out.println("Loan amount: "+amount);
        System.out.println("Loan interest rate: "+interestRate);
        System.out.println("____________________________");
    }

    void calculateLoan(int amount, double interestRate)
    {
        this.amount = amount;
        this.interestRate = interestRate;
        System.out.println("Loan amount: "+amount);
        System.out.println("Loan interest rate: "+interestRate);

    }
}


public class Q15Calculator {

    public static void main(String[] args) {

        LoanCalculator lc = new LoanCalculator();
        lc.calculateLoan(29000);
        lc.calculateLoan(17000, 7.99);
    }

}
```

```
<terminated> Q15Calculator [Java Application] C:\Users\admin\.p2\pool\
Loan amount: 29000
Loan interest rate: 0.0
____________________________
Loan amount: 17000
Loan interest rate: 7.99
```

Q16.

```java
package assignment;

class Hospital
{
    void emergencyService()
    {
        System.out.println("Emergency services");
    }
}

class CityHospital extends Hospital
{

    void emergencyService()
    {
        super.emergencyService();
        System.out.println("Emergency services from City Hospital");
    }
}

public class Q16Hospital {

    public static void main(String[] args) {

        CityHospital ch = new CityHospital();
        ch.emergencyService();
    }

}
```

<terminated> Q16Hospital [Java Application] C:\Users\admin\.p2\pool\p

```
Emergency services
Emergency services from City Hospital
```

Q17.

```java
package assignment;

abstract class Employee
{
    String empName;
    int empId;

    Employee(String name, int id)
    {
        this.empName=name;
        this.empId=id;
    }

    abstract void calculateSalary();

    void employeeDetails()
    {
        System.out.println("Employee name is: "+empName);
        System.out.println("Employee Id is: "+empId);
    }
}
class FullTimeEmployee extends Employee
{

    FullTimeEmployee(String name, int id)
    {
        super(name, id);

    }
    void calculateSalary()
    {
        System.out.println("Salary is: Per hour salary * 8hours");
    }
}

class PartTimeEmployee extends Employee
```

<terminated> Q17Abstract [Java Application] C:\Users\admin\.p2\pool\p

```
Employee name is: Bharat
Employee Id is: 780000
Salary is: Per hour salary * 8hours
Employee name is: Bharat BJ
Employee Id is: 7800
Salary is: Per hour salary * 4hours
```

```java
    }
    void calculateSalary()
    {
        System.out.println("Salary is: Per hour salary * 8hours");
    }
}

class PartTimeEmployee extends Employee
{
    PartTimeEmployee(String name, int id)
    {
        super(name, id);

    }

    void calculateSalary()
    {
        System.out.println("Salary is: Per hour salary * 4hours");
    }
}


public class Q17Abstract {

    public static void main(String[] args)
    {
        Employee e = new FullTimeEmployee("Bharat", 780000);
        e.employeeDetails();
        e.calculateSalary();

        Employee e1 = new PartTimeEmployee("Bharat BJ", 7800);
        e1.employeeDetails();
        e1.calculateSalary();
    }
}
```

```
<terminated> Q17Abstract [Java Application] C:\Users\admin\.p2\pool\p
Employee name is: Bharat
Employee Id is: 780000
Salary is: Per hour salary * 8hours
Employee name is: Bharat BJ
Employee Id is: 7800
Salary is: Per hour salary * 4hours
```

Q18.

```java
package assignment;

interface Transport
{
    void booking();
}

class Bus implements Transport
{
    public void booking()
    {
        System.out.println("This is bus service");
    }
}

class Flight implements Transport
{
    public void booking()
    {
        System.out.println("This is flight service");
    }

}

public class Q18Interface {

    public static void main(String[] args) {

        Transport tr = new Bus();
        tr.booking();
        Transport tr1 = new Flight();
        tr1.booking();

    }
```

```
<terminated> Q18Interface [Java Application] C:\Users\admin\.p2\pool\p
This is bus service
This is flight service
```

Q19.

```java
package assignment;

class Camera{
    void capture()
    {
        System.out.println("This is a camera");
    }
}

class DSLCamera extends Camera
{
    void capture()
    {
        System.out.println("This is a DSLCamera");
    }
}

public class Q19Polymor {

    public static void main(String[] args) {

        Camera cr = new DSLCamera();
        cr.capture();

    }

}
```

<terminated> Q19Polymor [Java Application] C:\Users\admin\.p2\pool\p

```
This is a DSLCamera
```

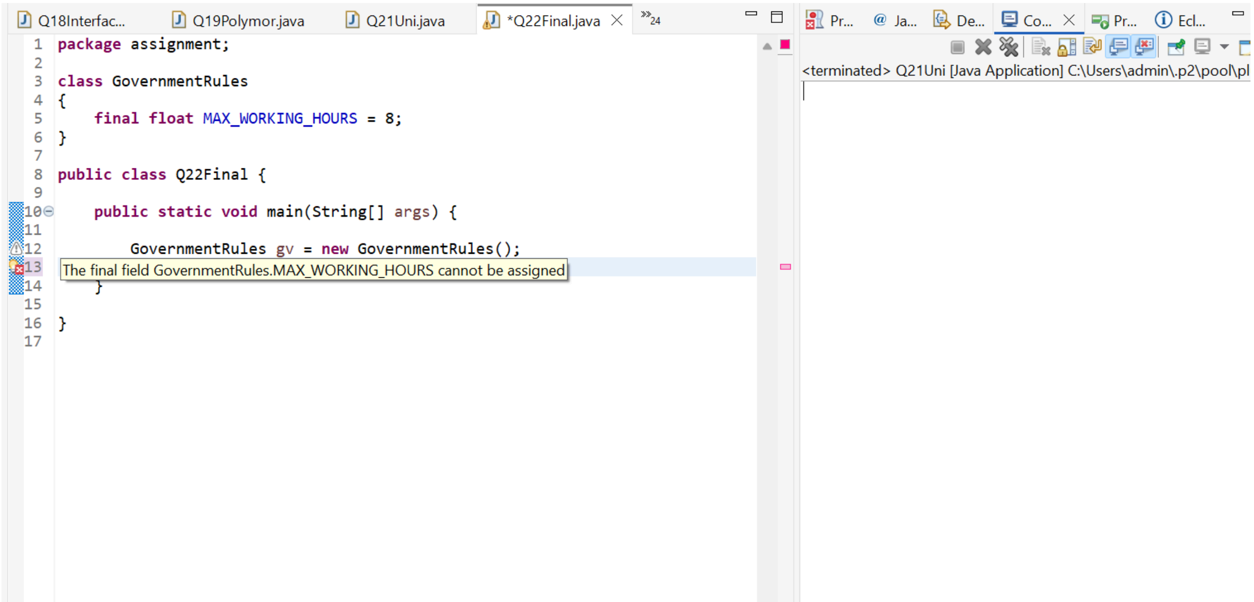
Q21.

```java
package assignment;

class University{
    static String country = "India";
    String universityName;

    University(String universityName)
    {
        this.universityName= universityName;
    }

    void display()
    {
        System.out.println("Country: "+country);
        System.out.println("University: "+universityName);
        System.out.println("_______________");

    }

}

public class Q21Uni {

    public static void main(String[] args) {

        University uni = new University("NIT");
        University uni1 = new University("IIT");
        uni.display();
        uni1.display();

    }

}
```

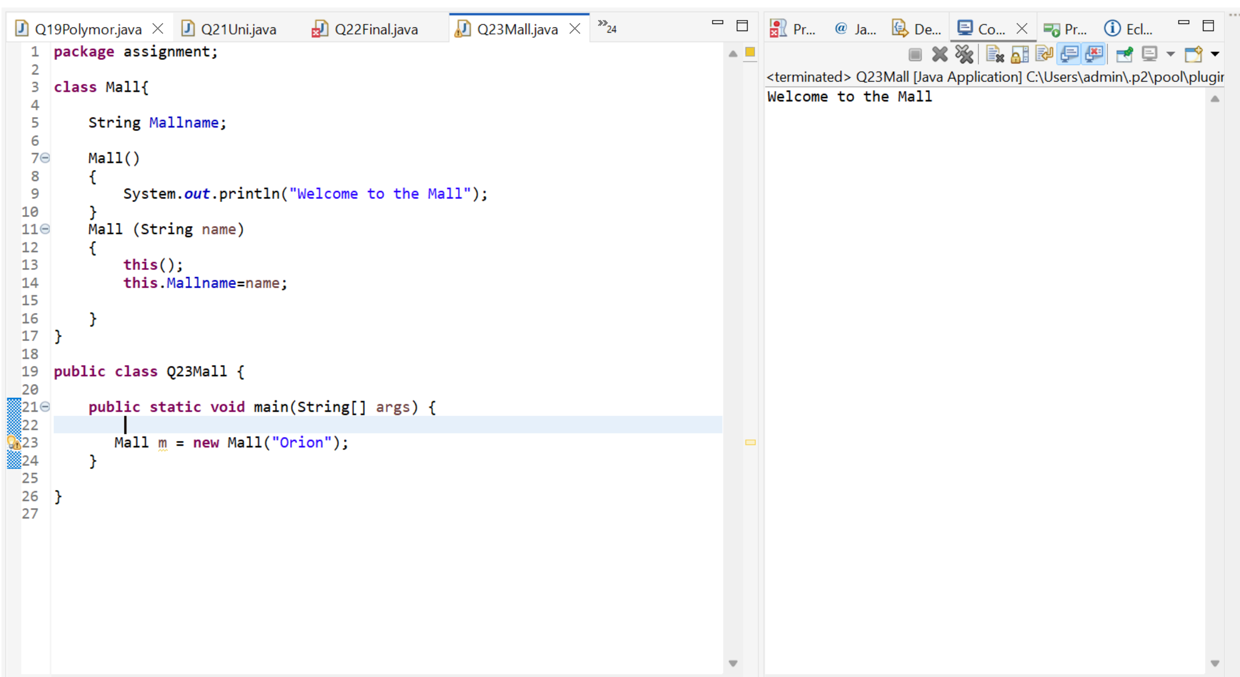
<terminated> Q21Uni [Java Application] C:\Users\admin\.p2\pool\plugin:

```
Country: India
University: NIT
_______________
Country: India
University: IIT
_______________
```

Q22.


Q18Interfac...
Q19Polymor.java
Q21Uni.java
*Q22Final.java
package assignment;
class GovernmentRules
{
final float MAX_WORKING_HOURS = 8;
}
public class Q22Final {
public static void main(String[] args) {
GovernmentRules gv = new GovernmentRules();
The final field GovernmentRules.MAX_WORKING_HOURS cannot be assigned
}
}
<terminated> Q21Uni [Java Application] C:\Users\admin\.p2\pool\pl

Q23.


Q19Polymor.java
Q21Uni.java
Q22Final.java
Q23Mall.java
package assignment;
class Mall{
String Mallname;
Mall()
{
System.out.println("Welcome to the Mall");
}
Mall (String name)
{
this();
this.Mallname=name;
}
}
public class Q23Mall {
public static void main(String[] args) {
Mall m = new Mall("Orion");
}
}
<terminated> Q23Mall [Java Application] C:\Users\admin\.p2\pool\plugir
Welcome to the Mall

Q24.

Q21Uni.java Q22Final.java Q23Mall.java Q24Shape.java

```java
package assignment;

class ShapeQ{
    float length;

    void square()
    {
        double area = length * length;
        System.out.println("Area of Square: " + area);
    }

    void rectangle(double breadth)
    {
        double area = length * breadth;
        System.out.println("Area of Rectangle: " + area);
    }

    void circle()
    {
        double area = 3.14 * length * length;
        System.out.println("Area of Circle: " + area);
    }
}

public class Q24Shape {

    public static void main(String[] args) {

        ShapeQ s = new ShapeQ();
        s.length = 4;

        s.square();
        s.rectangle(5.5);
        s.circle();

    }
```

<terminated> Q24Shape [Java Application] C:\Users\admin\.p2\pool\plu

```
Area of Square: 16.0
Area of Rectangle: 22.0
Area of Circle: 50.24
```

Writable Smart Insert 19 : 6 : 330

Q25.

Q22Final.java Q23Mall.java Q24Shape.java Q25School.java

```java
package assignment;

class School{
    String name;

    School()
    {
        name = "St. Xavier's School";
        System.out.println("The school name is: "+ name);
    }

    void display()
    {
        System.out.println("This School is based out of Kolkata");
    }
}

public class Q25School {

    public static void main(String[] args) {

        School s = new School();
        s.display();
    }

}
```

<terminated> Q25School [Java Application] C:\Users\admin\.p2\pool\plu

```
The school name is: St. Xavier's School
This School is based out of Kolkata
```

Q26.

```java
package assignment;

class School2
{
	String name;
	String address;
	String strength;

	School2(String name, String address){
		this.name=name;
		this.address= address;
	}

	School2(String name, String address, String strength){
		this.name=name;
		this.address=address;
		this.strength=strength;
	}

	void display()
	{
		System.out.println("Name is: "+name);
		System.out.println("Address is: "+address);
		System.out.println("Strength is: "+strength);
		System.out.println("__________________");
	}
}

public class Q26School {

	public static void main(String[] args) {
		School2 sc1 = new School2("School1", "Indranagar");
		sc1.display();
		School2 sc2 = new School2("School2","Rammurthynagar","Competative");
		sc2.display();
```

<terminated> Q26School [Java Application] C:\Users\admin\.p2\pool\plu

```
Name is: School1
Address is: Indranagar
Strength is: null
__________________
Name is: School2
Address is: Rammurthynagar
Strength is: Competative
__________________
```